

Spin Path Integrals and the Wess-Zumino-Witten Term

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In these notes I will derive a general coherent state path integral then specialize it for a single QM spin, taking primarily from Sachdev [3] with additional information from Atland [1] and Fradkin [2]. I found Sachdev's derivation to be elegant and easily followable (which I reiterate and expand on details), and Fradkin's continuations of the derivation were fitting.

Coherent State Path Integral Derivation

Assuming a single site, I will omit indices, and additionally take a general Hamiltonian of operators $H(\hat{\mathbf{S}})$. The coherent states $\{|\mathbf{n}\rangle\}$ are an ∞ -set of states labelled by a continuous vector $|\mathbf{n}\rangle$ that are normalized, non-orthogonal, and over-complete:

$$\langle \mathbf{n} | \mathbf{n} \rangle = 1, \quad \langle \mathbf{n} | \mathbf{n}' \rangle \neq 1, \quad \int d\mu(\mathbf{n}) |\mathbf{n}\rangle \langle \mathbf{n}| = 1, \quad (1)$$

where $\int d\mu(\mathbf{n})$ is a group integral over the Haar measure.¹ The fact that this integral is proportional to the identity is a consequence of Schur's Lemma $\diamond$:

$\diamond$ An operator $\hat{A}$ is proportional to the unit identity iff $\hat{A}$ commutes with all the representation matrices of an irreducible group representation.

The defining properties of the Haar measure provide this commutativity; take two group representations g, h , then

$$h \left(\int dg |g\rangle \langle g| \right) = \int dg |hg\rangle \langle g| \xrightarrow{\text{Haar}} \int dg |hh^{-1}g\rangle \langle h^{-1}g| = \left(\int dg |g\rangle \langle g| \right) h, \quad (2)$$

which follows because the Haar measure is invariant under left and right multiplication of fixed group elements. So, the integral commutes with all other representations, and is then proportional to the unit identity by Schur's Lemma. $\square$

Continuing, the property that defines $|\mathbf{n}\rangle$ as coherent states is the choice $\langle \mathbf{n} | \hat{\mathbf{S}} | \mathbf{n} \rangle = \mathbf{n}$, and I assume that $H(\hat{\mathbf{S}})$ is arranged such that $\langle \mathbf{n} | H(\hat{\mathbf{S}}) | \mathbf{n} \rangle = H(\mathbf{n})$.² I now have everything to construct the path integral, so define an imaginary time interval $0 \leq \tau \leq \beta$ into N_τ steps of length $\delta\tau$ such that $N_\tau \delta\tau = \beta$ and the partition function can be written

$$\mathcal{Z} = \text{Tr} e^{-H(\hat{\mathbf{S}})/T} = \lim_{N_\tau \rightarrow \infty, \delta\tau \rightarrow 0} \text{Tr} \left(e^{-\delta\tau H(\hat{\mathbf{S}})} \right)^{N_\tau}. \quad (3)$$

Between each adjacent i , I will insert the completeness relation (1). To order $\hat{\mathcal{O}}(\Delta\tau)$, I find

$$\begin{aligned} \langle \mathbf{n}(\tau) | e^{-\delta\tau H(\hat{\mathbf{S}})} | \mathbf{n}(\tau - \delta\tau) \rangle &\approx \langle \mathbf{n}(\tau) | (1 - \delta\tau H(\hat{\mathbf{S}})) | \mathbf{n}(\tau - \delta\tau) \rangle \\ &= 1 - \delta\tau \langle \mathbf{n}(\tau) | \partial_\tau \mathbf{n}(\tau) \rangle - \delta\tau H(\mathbf{n}) \\ &\approx \exp(-\delta\tau \langle \mathbf{n}(\tau) | \partial_\tau \mathbf{n}(\tau) \rangle - \delta\tau H(\mathbf{n})) \end{aligned} \quad (4)$$

after taking the linear approximation of $|\mathbf{n}(\tau - \Delta\tau)\rangle$. The partition function (3) can now be written as the path integral

$$\begin{aligned} \mathcal{Z} &= \int_{\mathbf{n}(0)=\mathbf{n}(\beta)} \mathcal{D}\mathbf{n}(\tau) \exp \left(- \int_0^\beta d\tau \langle \mathbf{n}(\tau) | \partial_\tau \mathbf{n}(\tau) \rangle - \int_0^\beta d\tau H(\mathbf{n}(\tau)) \right) \\ &\equiv \int_{\mathbf{n}(0)=\mathbf{n}(\beta)} \mathcal{D}\mathbf{n}(\tau) \exp \left(-S_B - \int_0^\beta d\tau H(\mathbf{n}(\tau)) \right), \end{aligned} \quad (5)$$

¹The Haar measure assigns an invariant volume to locally compact topological groups, and consequently defines an integral for functions on these groups.

²For the boson hubbard model, this is equivalent to normal-ordering.

where I have defined the "Berry phase" term

$$S_B = \int_0^\beta d\tau \langle \mathbf{n}(\tau) | \partial_\tau \mathbf{n}(\tau) \rangle, \quad (6)$$

which is purely imaginary and represents the overlap between two coherent states at infinitesimally separated times.

Berry Phase - Spin

Let $\hat{\mathbf{S}}$ in the above derivation be the Pauli vector in representation S , and demand $|\mathbf{n}|^2 = 1$ (so the $|\mathbf{n}\rangle$ are labelled by points on the unit sphere).³ Define the reference state to be $|\uparrow\rangle$, from which all other coherent states can be obtained through an $SU(2)$ rotation (action of group representations):

$$|\mathbf{n}\rangle = \exp\left(z\hat{S}^+ - z^*\hat{S}^-\right) |\uparrow\rangle = \exp\begin{pmatrix} 0 & z \\ -z^* & 0 \end{pmatrix} |\uparrow\rangle \equiv U |\uparrow\rangle, \quad (7)$$

where in spherical coordinates, $z = -(\theta/2)e^{-i\phi}$ works.

Developing the Berry phase term (6), note that

$$\partial_\tau |\mathbf{n}(\tau)\rangle = \partial_\tau \left(\exp\left(z\hat{S}^+ - z^*\hat{S}^-\right) |\uparrow\rangle \right), \quad (8)$$

which is not necessarily trivial, as

$$\partial_\tau(e^{\hat{\mathcal{O}}}) = \int_0^1 du e^{\hat{\mathcal{O}}(1-u)} \partial_\tau(\hat{\mathcal{O}}) e^{\hat{\mathcal{O}}u}, \quad (9)$$

but does give the intuitive result $\partial_t(\hat{\mathcal{O}})e^{\hat{\mathcal{O}}}$ if $\mathcal{O}$ commutes with its derivative. With this in mind, (6) can be written

$$\begin{aligned} S_B &= \int_0^\beta d\tau \langle \uparrow | e^{-\hat{\mathcal{O}}} \int_0^1 du e^{\hat{\mathcal{O}}(1-u)} \partial_\tau(\hat{\mathcal{O}}) e^{\hat{\mathcal{O}}u} | \uparrow \rangle \\ &= \int_0^\beta d\tau \int_0^1 du \langle \uparrow | e^{-\hat{\mathcal{O}}u} \partial_\tau(\hat{\mathcal{O}}) e^{\hat{\mathcal{O}}u} | \uparrow \rangle \\ &= \int_0^\beta d\tau \int_0^1 du \langle \mathbf{n}(\tau, u) | \left(\partial_\tau z \hat{S}^+ - \partial_\tau z^* \hat{S}^- \right) | \mathbf{n}(\tau, u) \rangle, \end{aligned} \quad (10)$$

where I have defined

$$|\mathbf{n}(\tau, u)\rangle \equiv \exp\left[u(z\hat{S}^+ - z^*\hat{S}^-)\right] |\uparrow\rangle. \quad (11)$$

Note the following properties of (11),

$$\mathbf{n}(\tau, u=1) = \mathbf{n}(\tau), \quad \mathbf{n}(\tau, u=0) = \hat{e}_3, \quad (12)$$

so $\mathbf{n}(\tau, u)$ moves with u along the great circle between $\mathbf{n}(\tau, 1)$ and the north pole $\mathbf{n}(\tau, 0)$. Making the $SU(2)$ symmetry more manifest, define the quantities $\hat{\mathcal{S}}, W$ with the forms

$$\hat{\mathcal{S}} = \begin{pmatrix} \hat{S}^z & \hat{S}^- \\ \hat{S}^+ & -\hat{S}^z \end{pmatrix}, \quad \langle \mathbf{n}(\tau, u) | \hat{\mathcal{S}} | \mathbf{n}(\tau, u) \rangle = S \cdot W(\tau, u), \quad (13)$$

so $W = \mathbf{n} \cdot \boldsymbol{\sigma}$ is identified as the mapping of the coherent (vector) basis to the Pauli (matrix) basis. The following relations follow,

$$\langle \mathbf{n}(\tau, u) | \hat{S}^+ | \mathbf{n}(\tau, u) \rangle = S \hat{W}_{21}, \quad \langle \mathbf{n}(\tau, u) | \hat{S}^- | \mathbf{n}(\tau, u) \rangle = S \hat{W}_{12}, \quad (14)$$

so now (10) can be written

$$S_B = S \int_0^\beta d\tau \int_0^1 du (\partial_\tau z W_{21}(\tau, u) - \partial_\tau z^* W_{12}(\tau, u)). \quad (15)$$

³In this case, $d\mu(\mathbf{n}) = \left(\frac{2S+1}{4\pi}\right) \delta^2(|\mathbf{n}|^2 - 1) d^3n$ is the measure, where the normalization is not physically relevant for an action.

The idea is to now perform an I.B.P on τ , where since the integrand is periodic over the range of integration, the surface term vanishes, giving

$$\begin{aligned} S_B &= -S \int_0^\beta d\tau \int_0^1 du (z(\tau) \partial_\tau W_{21}(\tau, u) - z^*(\tau) \partial_\tau W_{12}(\tau, u)) \\ &= -S \int_0^\beta d\tau \int_0^1 du \text{Tr} \left[\begin{pmatrix} 0 & z \\ -z^* & 0 \end{pmatrix} \partial_\tau W(\tau, u) \right]. \end{aligned} \quad (16)$$

This integrand can be made to look much prettier; note that (7) implies

$$\begin{aligned} U(\tau) \sigma_z U^\dagger(\tau) &= \sigma_z + \left[\begin{pmatrix} 0 & z \\ -z^* & 0 \end{pmatrix}, \sigma_z \right] + \mathcal{O}(|z|^2) = \begin{pmatrix} 1 & -2z \\ -2z^* & -1 \end{pmatrix} + \mathcal{O}(|z|^2) \\ &= \begin{pmatrix} 1 & \theta(\cos \phi - i \sin \phi) \\ \theta(\cos \phi + i \sin \phi) & -1 \end{pmatrix} \\ &= \begin{pmatrix} n_z & n_x - i n_y \\ n_x + i n_y & -n_z \end{pmatrix} \\ &= W(\tau), \end{aligned} \quad (17)$$

where I first used the Baker-Campbell-Hausdorff identity, truncating the series to first order in z (or θ), then I used $z = -(\theta/2)e^{-i\phi}$ (with $1 \sim \cos \theta$ & $\theta \sim \sin \theta$). It follows, then, that

$$\begin{aligned} -\frac{1}{2} W(\tau, u) \partial_u W(\tau, u) &= -\frac{1}{2} \begin{pmatrix} 1 & -2uz \\ -2uz^* & -1 \end{pmatrix} \begin{pmatrix} 0 & -2z \\ -2z^* & 0 \end{pmatrix} = -\frac{1}{2} \begin{pmatrix} 4u|z|^2 & -2z \\ 2z^* & 4u|z|^2 \end{pmatrix} \\ &= \begin{pmatrix} 0 & z \\ -z^* & 0 \end{pmatrix} + \mathcal{O}(|z|^2), \end{aligned} \quad (18)$$

and the Berry phase (16) can be written

$$S_B = \frac{S}{2} \int_0^\beta d\tau \int_0^1 du \text{Tr} [W(\tau, u) \partial_u W(\tau, u) \partial_\tau W(\tau, u)]. \quad (19)$$

Using the fact that $W(\tau, u) = \mathbf{n}(\tau, u) \cdot \sigma$,

$$\begin{aligned} S_B &= \frac{S}{2} \int_0^\beta d\tau \int_0^1 du \text{Tr} [(\mathbf{n}(\tau, u) \cdot \sigma) (\partial_u \mathbf{n}(\tau, u) \cdot \sigma) (\partial_\tau \mathbf{n}(\tau, u) \cdot \sigma)] \\ &= \frac{S}{2} \int_0^\beta d\tau \int_0^1 du n^i \partial_u n^j \partial_\tau n^k \text{Tr} [\sigma^i \sigma^j \sigma^k] \\ &= \frac{S}{2} \int_0^\beta d\tau \int_0^1 du n^i \partial_u n^j \partial_\tau n^k (2i\epsilon^{ijk}), \end{aligned} \quad (20)$$

$$\boxed{S_B = iS \int_0^\beta d\tau \int_0^1 du \mathbf{n} \cdot (\partial_u \mathbf{n} \times \partial_\tau \mathbf{n})} \quad (21)$$

which is quite elegant.⁴ This expression is called a symplectic 2-form in the mathematical context, but is also known as the Wess-Zumino action, or simply the Berry phase in the realm of physics.

Interpretation

The function $\mathbf{n}(\tau, u)$ is a mapping from the rectangle $\tau \in [0, \beta]$, $u \in [0, 1]$ to the unit sphere. As $\mathbf{n}(\tau, u)$ moves along the sphere it drags along with it a string which connects to the north pole, and the area of the region swept out by the string in some interval $\tau \rightarrow \tau + d\tau$ is

$$dA[\Omega^+] = d\tau \left(\int_0^1 du \mathbf{n} \cdot (\partial_u \mathbf{n} \times \partial_\tau \mathbf{n}) \right), \quad (22)$$

⁴To evaluate the trace I used $\sigma^i \sigma^j = \delta^{ij} I + i\epsilon^{ijk} \sigma^k$ and the fact that $\text{Tr}[I] = 2$ and $\text{Tr} \sigma^i = 0$.

where the contribution to the Berry phase is simply iS times this area. Periodic boundary conditions imply that $\mathbf{n}(\tau, u)$ taken over the full region $\tau \in [0, \beta]$ traces out a closed, oriented curve $\Gamma = \partial\Omega^+$ that bisects the unit sphere into hemispheres $\Omega^\pm$ (Fig. 1).

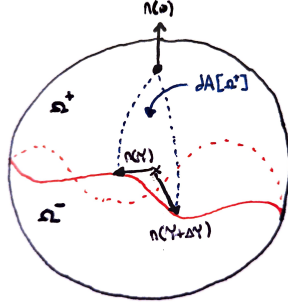


Figure 1: Vector $\mathbf{n}$ sweeps a closed path Γ (red) that bisects the unit sphere into hemispheres $\Omega^\pm$, which in the process drags a string (blue) which sweeps an area element (22) contributing to the Berry phase.

Since my gauge choice is the north pole, $\mathbf{n}(\tau, 0) = (0, 0, 1)$, the corresponding Berry phase has the form $S_B = iS \times A[\Omega^+]$. There is an issue, however, as the "inside" of a closed loop is ill-defined, leaving the area uncertain modulo 4π : $A[\Omega^+] = 4\pi - A[\Omega^-]$. The net contribution of this uncertainty to the path integral (5) is $\exp(i(4\pi S))$, which is alleviated by the restriction of S being integer or half-integer.

References

- [1] Alexander Altland and Ben D. Simons. *Condensed Matter Field Theory*. Cambridge University Press, 2023.
- [2] Eduardo Fradkin. *Field Theories of Condensed Matter Physics*. Cambridge University Press, 2 edition, 2015.
- [3] Subir Sachdev. *Quantum Phases of Matter*. Cambridge University Press, 2023.